

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

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Hand over the address, not the sentence.

1 · THE OBJECT AND ITS INVARIANT

The bytes are their own name.

```
{ kind: "primary",
  cell: "defi.zb493.xuqA.zcb5f", // 64-bit cell ≈9.55 m
  band: "copdem30m.elevation_mean",
  tslot: 0, // valid time
  value: 918.0, unit: "m", confidence: 0.95,
  sources: [{ scheme: "open_meteo",
    captured_at: "2021-04-30" }],
  derivation: { fn_key: "open_meteo_copdem90m@1" },
  schema_cid: "d24rgwlq...",
  signer: ed25519:777er3yi...,
  signed_at: "2026-05-28T19:54:32Z" // transaction time }
```

ENCODE

emem-CBOR · 509 bytes
ad 64 6b 69 6e 64 67 70 72 69 6d 61 72 79 64 63 65 6c 6c ...

HASH

BLAKE3-256 → base32

NAME

yqbo1geoycqkvj3zkxukb4bjw4odhpwvfzqo3fbgwf4spk45zala

→

emem:fact:defi.zb493.xuqA.zcb5f:yqbo1geoycqkvj3zkxukb4bjw4odhpwvfzqo3fbgwf4spk45zala

token = where + which bytes · 84 characters

live on emem.dev

Same bytes → same name. Anyone re-derives it with a stock BLAKE3; no emem code, no registry.

One value changed → a different name. 918.0 → 918.1 gives vines361...

Wrong place → refused. This fact cited under another cell returns HTTP 409, never a silent wrong answer.

Deterministic CBOR in struct-declaration order, NaN and −0 canonicalised (not RFC 8949 key-sorted). The fact carries its signer and signing time; the ed25519 signature covers a Merkle root of fact CIDs. A CID therefore names *one signed attestation*: two attesters of the same value mint different CIDs and meet at the key (cell, band, tslot).

3 · FOUND IN PRODUCTION, NOT STAGED

One place. Two answers. Both still checkable.

Token signed 28 May 2026
918.0 m
Bengaluru · copdem30m.elevation_mean
source open_meteo · Copernicus DEM 90 m
emem:fact:defi.zb493.xuqA.zcb5f:yqbo1geo...
re-hash 29 Sep 2026: MATCH · 509 bytes

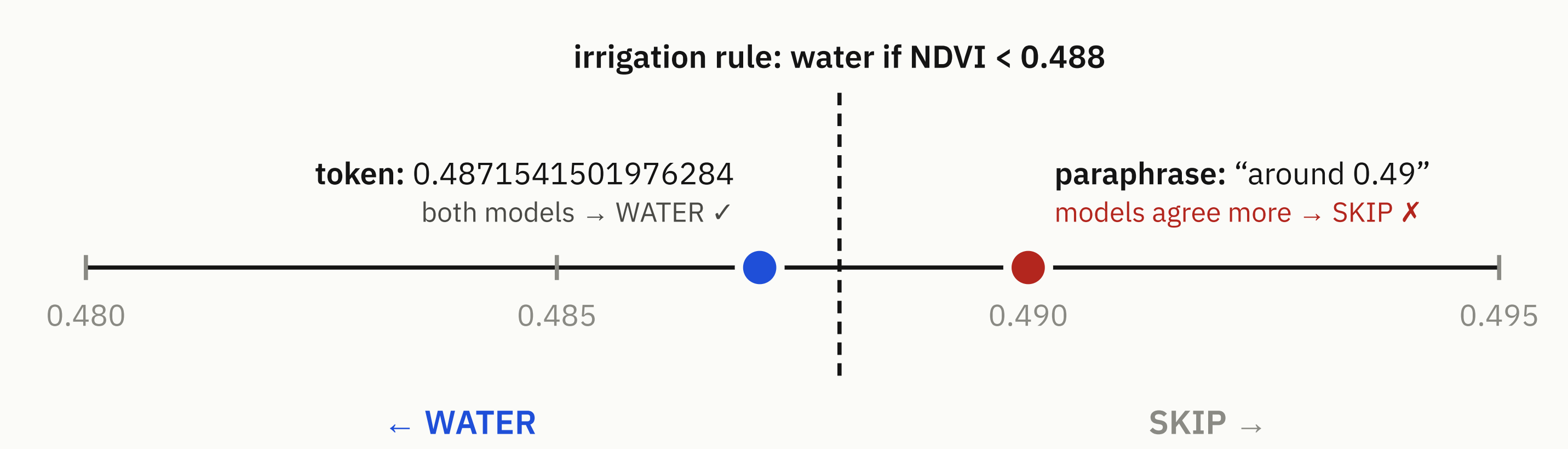
→
upstream provider changed

Live answer, signed 28 Sep 2026
915.07 m
same cell · same band
source copernicus.dem.30m.aws · COG tile N12 E077
emem:fact:defi.zb493.xuqA.zcb5f:jxzmvmom...
re-hash 29 Sep 2026: MATCH · 718 bytes

An agent that kept the sentence “*elevation* ≈ 918 m” cannot tell that the answer moved, or why. An agent that kept the **token** still gets 918.0, with the dataset, recipe and signing time that produced it — and can put both answers side by side.

4 · THE FINDING

A paraphrase made two models agree — and both chose wrong.



One real Lahaul NDVI fact, Gemma-4-12B and Qwen2.5-7B. The lossy memory raised cross-model agreement while carrying the wrong decision.

Pre-registered test — accuracy vs agreement after a shared summary	
full context (control)	72/72 correct · 36/36 agree
compaction, free reasoning	20/72 · 15/36 p = 0.109
compaction under pressure	0/72 · 3/36 p = 0.035

Agreement between agents is not evidence of correctness. It is what a shared lossy summary produces. Fisher one-sided; two open models on one host; six scorer bugs found and fixed before the verdict.

6 · OVER FOUNDATION-MODEL EMBEDDINGS AND FIELDS

From a point to a field, same rule.

fact	one signed observation <cell64>:<fact_cid>	BYTE-IDENTICAL
raster	a Sentinel-2 field over an AOI, f32 grid <aoi>:<band>:<tslot>:<derivation>	DERIVATION-BOUND
cube	that field through time: signed manifest of rasters <aoi>:<band>:<lo>.. <hi>:<derivation>< td=""><td>DERIVATION-BOUND</td></hi>:<derivation><>	DERIVATION-BOUND
rasteriset	2–64 fields bound as one model input	DERIVATION-BOUND
bundle entity	a set of facts · a shared name for a thing	ANCHOR

A field token binds a signed derivation: AOI, source scene, recipe, time and the artifact’s hash. The model receives **a field, not a sentence about a field**; the pixels can be evicted and rebuilt from the recipe. **Embeddings** (live: TESSERA, 128-D; earlier Clay v1.5, Prithvi-EO-2.0, Galileo) are signed as ordinary fact values labelled model_output — a *learned representation, not a measurement*. The embedding is not the memory; the addressable observation around it is.

8 · REPRODUCE THIS — NOW, ON YOUR PHONE OR LAPTOP

```
# pip install blake3          — no emem code involved
import blake3, base64, urllib.request as u
cid = "yqbo1geoycqkvj3zkxukb4bjw4odhpwvfzqo3fbgwf4spk45zala"
r = u.Request(f"https://emem.dev/v1/facts/{cid}", headers={f"accept": "application/cbor"})
body = u.urlopen(r).read()          # 509 bytes → value 918.0 m
name = base64.b32encode(blake3.blake3(body).digest()).decode().lower().rstrip("=")
assert name == cid                  # the bytes are their own name

# give your agent the same memory:
claude mcp add --transport http emem https://emem.dev/mcp
```

MCP + REST from one Rust binary · 18 core / 114 MCP tools · 216 agents and 8,427 signed notes in the public collaboration log · mutual log witnessing with an independent operator (geo.qa). Numbers: emem docs/benchmarks, how-emem-compares, paper-section-statistics; single node, open models, no independent replication yet.

emem names each physical observation by a hash of its own signed bytes. An agent passes the **name**. Any receiver fetches the bytes, **re-hashes them**, and **verifies the signature offline** — without trusting the sender, its model, or the server.

2 · THE RECEIVER

Resolve. Don’t trust.

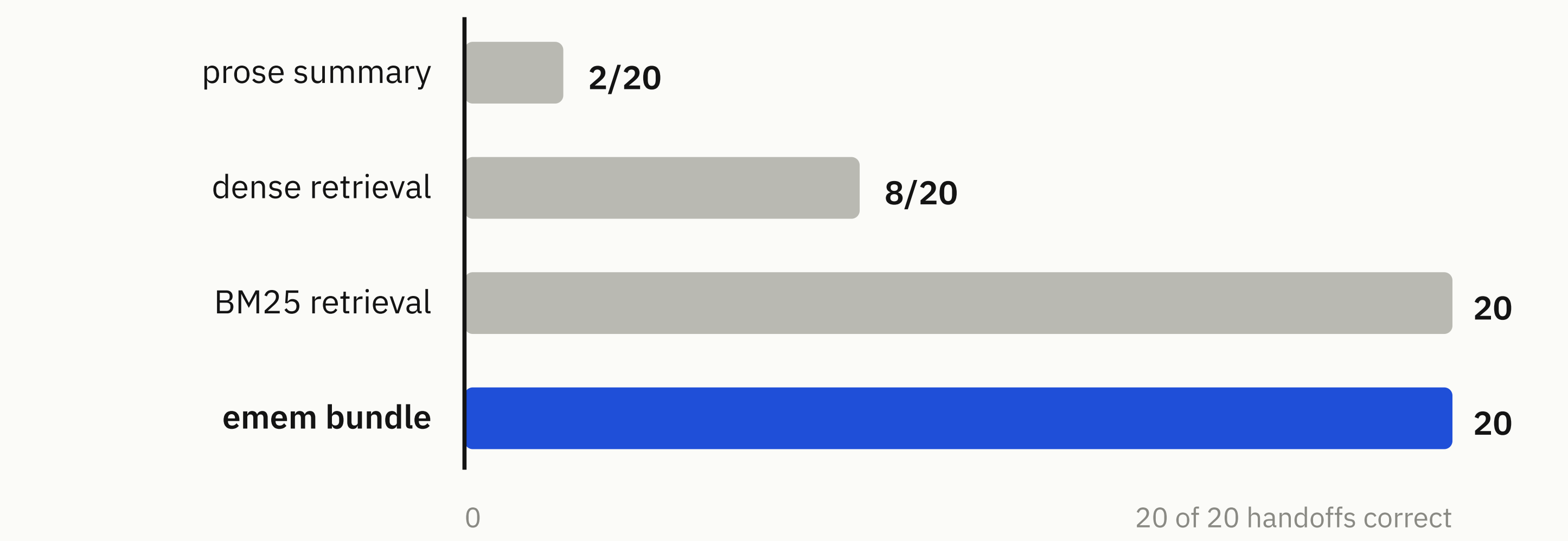
- 1 Resolve** the token → the exact bytes GET /v1/facts/<cid>
- 2 Re-hash** them yourself. Does the name match?
- 3 Bind the place**: the cell in the token must equal the cell inside the signed fact.
- 4 Verify the receipt** offline: ed25519 over a domain-separated, length-prefixed preimage. **0.13 ms** p50.
- 5 Check history**: RFC 6962-style log with BLAKE3; **2.54 M entries** (29 Sep), co-signed by an independent operator, geo.qa.

CROSSES THE BOUNDARY	DOES NOT NEED TO
84 characters. Or one emem:bundle: handle — 38 chars for up to 256 facts.	the sender’s prompt, context window, model, database, or honesty.
A proof-less receipt signs an explicit ABSENT marker, so stripping a proof is detectable. An independent audit rejected 5/5 tampers, incl. a version downgrade.	

Noticed by an external auditor in August; independently re-verified by us on 29 Sep 2026 using only the blake3 and cbor2 reference libraries. Superseded facts are never deleted; corrections are new facts with new names.

5 · HANDOFF BETWEEN AGENTS

What Agent B gets right, by what Agent A handed over.



multiagent_v2, n = 20 per arm, Gemma ↔ Qwen

Long run — 599 turns, n = 1,624 per arm	within 1% · citable
emem bundle	100% · 100%
BM25 retrieval	35% · 0%

BM25 ties on short handoffs — say it plainly. What it cannot give the next agent is a citation that re-checks. Over long runs it drifts; the addressed bundle stays exact and every answer stays citable.

7 · BOUNDARY OF THE CLAIM

What holds. What doesn’t.

CHECKABLE	NOT PROVED
✓ byte integrity of a cited fact	✗ that the observation is true
✓ where and when it claims to be	✗ sensor calibration or model accuracy
✓ who signed it, and when	✗ that two sensors saw the same thing
✓ that history was not rewritten	✗ that a sentence cites the right band
✓ “nothing here” — absence is a signed fact	✗ anything in orbit — edge signing is simulated (0.17% of a 40 ms frame)
✓ disagreement: kept and scored, never averaged	
Our own scorecard	
A citation survives compaction where a paraphrase does not	supported
Beats dense retrieval	supported
Beats plain context · beats BM25	refuted
A single token saves context (it costs 9.5x; bundles do)	refuted
Beats peer agent-memory products	not tested
Open: machine-checked protocol proofs, full conformance vectors, hardware-attested devices, read federation.	



Repro scripts
every number on this poster



Verify in browser
emem.dev/verify



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